Combinatorics (COMB.)

Part 1: Enumeration

Author: Jaehoon Song *Release: 2025-08-07 22:22:04-04:00*

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Prepared by Jaehoon Song, TA@GT

1. **Rule of Product 1** (10 points)
The Hawaiian alphabet consists of 12 letters. How many six-character strings can be made using the Hawaiian alphabet?

Answer: 12^6. Label each letter in the Hawaiian alphabet as $1, 2, \ldots, 12$. Then, each string corresponds to an element in $\{1, \ldots, 12\}^6$.

$$\{1, \ldots, 12\} \times \{1, \ldots, 12\} \times \{1, \ldots, 12\} \times \{1, \ldots, 12\} \times \{1, \ldots, 12\} \times \{1, \ldots, 12\}$$

$\square$

2. **Rule of Product 2** (10 points)
How many $2n$-digit positive integers can be formed if the digits in odd positions (counting the rightmost digit as position 1) must be odd and the digits in even positions must be even and positive?

Answer:

$\square$

3. **Rule of Product 3** (10 points)
Matt is designing a website authentication system. He knows passwords are most secure if they contain letters, numbers, and symbols. However, he doesn't quite understand that this additional security is defeated if he specifies in which positions each character type appears. He decides that valid passwords for his system will begin with three letters (uppercase and lowercase both allowed), followed by two digits, followed by one of 10 symbols, followed by two uppercase letters, followed by a digit, followed by one of 10 symbols. How many different passwords are there for his website system? How does this compare to the total number of strings of length 10 made from the alphabet of all uppercase and lowercase English letters, decimal digits, and 10 symbols?

Answer: The number of passwords with Matt's specified placements is $52^3 \cdot 10^3 \cdot 26^2 \cdot 10^2$. The number of strings made from all of the available characters is $(26 + 26 + 10 + 10)^{10} = 72^{10}$. The second answer is approximately 393,884 times larger than Matt's set of passwords.

$\square$

4. **Rule of Product 4** (10 points)
How many ternary strings of length $2n$ are there in which the zeroes appear only in odd-numbered positions?

Answer:

$\square$

5. **Rule of Product 5** (10 points)
How many strings of the form $l_1 l_2 d_1 d_2 d_3 l_3 l_4 d_4 l_5 l_6$ are there where for $1 \leq i \leq 6$, l_i is an uppercase letter in the English alphabet; for $1 \leq i \leq 4$, d_i is a decimal digit; l_2 is not a vowel (i.e., $l_2 \notin \{A, E, I, O, U\}$); and the digits d_1, d_2, d_3 are distinct (i.e., $d_1 \neq d_2 \neq d_3 \neq d_1$).

Answer: $26^5 \cdot (26 - 5) \cdot 10^2 \cdot 9 \cdot 8$. We have 26 choices for l_1, l_3, l_4, l_5, l_6, but only $26 - 5 = 21$ choices for l_2. We have $10 \cdot 9 \cdot 8 = P(10, 3)$ choices for d_1, d_2, d_3 since they form a permutation of 3 digits on the alphabet $\{0, 1, \ldots, 9\}$.

$\square$

6. **Enumeration[1] 1** (10 points)
Mrs. Steffen's third grade class has 30 students in it. The students are divided into three groups (numbered 1, 2, and 3), each having 10 students.

   a. The students in group 1 earned 10 extra minutes of recess by winning a class competition. Before going out for their extra recess time, they form a single file line. In how many ways can they line up?

      Answer:

      □

   b. When all 30 students come in from recess together, they again form a single file line. However, this time the students are arranged so that the first student is from group 1, the second from group 2, the third from group 3, and from there on, the students continue to alternate by group in this order. In how many ways can they line up to come in from recess?

      Answer:

      □

7. **Enumeration 2** (10 points)
In this exercise, we consider strings made from uppercase letters in the English alphabet and decimal digits. How many strings of length 10 can be constructed in each of the following scenarios?

   a. The first and last characters of the string are letters.

      Answer:

      □

   b. The first character is a vowel, the second character is a consonant, and the last character is a digit.

      Answer:

      □

   c. Vowels (not necessarily distinct) appear in the third, sixth, and eighth positions and no other positions.

      Answer:

      □

   d. Vowels (not necessarily distinct) appear in exactly two positions.

      Answer:

      □

   e. Precisely four characters in the string are digits and no digit appears more than one time.

      Answer:

      □

[1] In combinatorics, **enumeration** refers to the process of systematically counting the number of elements in a set or the number of ways a particular arrangement can be formed. It includes **permutations** (ordered selections with or without repetition) and **combinations** (unordered selections with or without repetition).

8. **Enumeration 3** (10 points)
A database uses 20-character strings as record identifiers. The valid characters in these strings are upper-case letters in the English alphabet and decimal digits. (Recall there are 26 letters in the English alphabet and 10 decimal digits.) How many valid record identifiers are possible if a valid record identifier must meet all of the following criteria:

- Letter(s) from the set $\{A, E, I, O, U\}$ occur in exactly three positions of the string.
- The last three characters in the string are distinct decimal digits that do not appear elsewhere in the string.

Answer: $\binom{17}{3} \cdot 5^3 \cdot 10 \cdot 9 \cdot 8 \cdot 28^{14}$. We first choose where we place the vowels. Since they cannot be placed in the last 3 positions, we have 17 available spots from which we choose a subset of size 3: $\binom{17}{3}$. Once the spots for the vowels are chosen, we have 5 choices for which vowel is placed in each of the 3 positions: 5^3. Then, we place 3 distinct decimal digits at the end of which there are $P(10, 3) = 10 \cdot 9 \cdot 8$ such strings. Lastly, we have 14 positions left to fill, and those 14 positions cannot contain vowels or any of the 3 decimal digits chosen previously: 28^{14}.

□

9. **Enumeration 4** (10 points)
Let X be the set of the 26 lowercase English letters and 10 decimal digits. How many X-strings of length 15 satisfy all of the following properties (at the same time)?

- The first and last symbols of the string are distinct digits (which may appear elsewhere in the string).
- Precisely four of the symbols in the string are the letter 't'.
- Precisely three characters in the string are elements of the set $V = \{a, e, i, o, u\}$ and these characters are all distinct.

Answer: $P(10, 2) \cdot \binom{13}{4} \cdot \left(\binom{9}{3} \cdot P(5, 3)\right) \cdot 30^6$. Place 2 distinct decimal digits at the first and last of the string, there are $P(10, 2)$ such strings. Choose where to place the letter 't'. Since the first and last positions are not available, 4 positions will be chosen from 13 available positions: $\binom{13}{4}$. Choose 3 positions for vowels (a, e, i, o, u) from the rest of 9 positions, then place 3 distinct vowels in the positions: $\binom{9}{3} \cdot P(5, 3)$. Lastly, we have 6 positions left to fill, and those 6 positions cannot contain vowels or 't' chosen previously: 30^6.

□

10. **Enumeration 5** (10 points)
A donut shop sells 12 types of donuts. A manager wants to buy six donuts, one each for himself and his five employees.

   a. Suppose that he does this by selecting a specific type of donut for each person. (He can select the same type of donut for more than one person.) In how many ways can he do this?

   Answer: 12^6 since each selection is a string of length 6 (the positions correspond to the employee that the donut will go to) from an alphabet of size 12 (the donut flavors).

   □

   b. How many ways could he select the donuts if he wants to ensure that he chooses a different type of donut for each person?

   Answer: $P(12, 6) = 12 \cdot 11 \cdot 10 \cdot 9 \cdot 8 \cdot 7$. These are strings of length 6 from an alphabet of size 12 that are also permutations.

   □

   c. Suppose instead that he wishes to select one donut of each of six different types and place them in the breakroom. In how many ways can he do this? (The order of the donuts in the box is irrelevant.)

   Answer: $\binom{12}{6}$. He is now selecting a subset of 6 donuts (because the donuts must be different) from a set of 12.

   □

11. **Enumeration 6** (10 points)
Twenty students compete in a programming competition in which the top four students are recognized with trophies for first, second, third, and fourth places.

   a. How many different outcomes are there for the top four places?

   Answer: $P(20, 4) = 20 \cdot 19 \cdot 18 \cdot 17$ since we are considering strings of length 4 (the position in the string corresponds to the trophy) from an alphabet of size 20 (the students in the competition).

   □

   b. At the last minute, the judges decide that they will award honorable mention certificates to four individuals who did not receive trophies. In how many ways can the honorable mention recipients be selected (after the top four places have been determined)? How many total outcomes (trophies plus certificates) are there then?

   Answer: $\binom{16}{4}$ because we are selecting a subset of 4 students from the set of 16 students who did not win a trophy. The total number of outcomes is $P(20, 4) \cdot \binom{16}{4}$.

   □

12. **Enumeration 7** (10 points)
An ice cream shop has a special on banana splits, and Xing is taking advantage of it. He's astounded at all the options he has in constructing his banana split:

- He must choose three different flavors of ice cream to place in the asymmetric bowl the banana split is served in. The shop has 20 flavors of ice cream available.
- Each scoop of ice cream must be topped by a sauce, chosen from six different options. Xing is free to put the same type of sauce on more than one scoop of ice cream.
- There are 10 sprinkled toppings available, and he must choose three of them to have sprinkled over the entire banana split.

a. How many different ways are there for Xing to construct a banana split at this ice cream shop?

Answer:

□

b. Suppose that instead of requiring that Xing choose exactly three sprinkled toppings, he is allowed to choose between zero and three sprinkled toppings. In this scenario, how many different ways are there for him to construct a banana split?

Answer:

□

13. **Enumeration 8 (Permutations with repetition)** (10 points)
For each word below, determine the number of rearrangements of the word in which all letters must be used.

a. OVERNUMEROUSNESSES
b. OPHTHALMOOTORHINOLARYNGOLOGY
c. HONORIFICABILITUDINITATIBUS (the longest word in the English language consisting strictly of alternating consonants and vowels 1/1)

Answer: (a) 18!/4!4!2!2!2!2!1!1!. There are 18! permutations of length 18. However, since the letter S is repeated 4 times, we must divide by the number of permutations of length 4 , and so on. Doing this for each letter in the word results in the answer. (b) 28!/7!3!3!2!2!2!2!2!2!1!1!1! (c) 27!/7!3!2!2!2!2!2!1!1!1!1!1!1!1!

□

14. **Enumeration 9 (Permutations with repetition)** (10 points)
How many ways are there to paint a set of 27 elements such that 7 are painted white, 6 are painted old gold, 2 are painted blue, 7 are painted yellow, 5 are painted green, and 0 are painted red?

Answer: $27!/(7! \cdot 6! \cdot 2! \cdot 7! \cdot 5! \cdot 0!)$. There are 27! permutations of elements. However, since 7 are painted white (repeated), we must divide by the number of permutations of length 7, and so on for the rest of them. Doing this for each letter in the word results in the answer. $27!/(7! \cdot 6! \cdot 2! \cdot 7! \cdot 5! \cdot 0!)$.

□

15. **Enumeration 10 (Combinations with repetition)** (10 points)
Suppose that a teacher wishes to distribute 25 identical pencils to Ahmed, Barbara, Casper, and Dieter such that Ahmed and Dieter receive at least one pencil each, Casper receives no more than five pencils, and Barbara receives at least four pencils. In how many ways can such a distribution be made?

Answer: $(22, 3) - (16, 3)$. We are distributing 25 identical objects (the pencils) into 4 distinguishable bins (the four students). We also have restrictions on the number of objects in each bin. There are $(25 + 4 - 1 - (1 + 1 + 4), 4 - 1) = (22, 3)$ ways to do this where Ahmed receives at least one, Dieter receives at least one, and Barbara receives at least four. We subtract out the ways in which Casper receives at least six, $(16, 3)$ to get the total number in which Casper receives at most five.

□

16. **Enumeration 11 (Combinations with repetition)** (10 points)
How many integer solutions are there to inequality $x_1 + x_2 + x_3 + x_4 + x_5 \leq 782$ provided that $x_1, x_2 > 0$, $x_3 \geq 0$, and $x_4, x_5 \geq 10$?

Answer: $(x_1 - 1) + (x_2 - 1) + (x_3) + (x_4 - 10) + (x_5 - 10) \leq 782 - 22 = 760$ $(x_1 - 1) + (x_2 - 1) + (x_3) + (x_4 - 10) + (x_5 - 10) \leq 782 - 22 = 760$ $\sum_{k=0}^{760} \binom{k+4}{4}$ or $(x_1 - 1) + (x_2 - 1) + (x_3) + (x_4 - 10) + (x_5 - 10) + x_6 = 760$ $(760 + 5, 5) = (765, 5)$

□

17. **Combinations: Binomial Coefficient 1** (10 points)
The sport of korfball is played by teams of eight players. Each team has four men and four women on it. Halliday High School has seven men and 11 women interested in playing korfball. In how many ways can they form a korfball team from their 18 interested students?

Answer: $\binom{7}{4} \cdot \binom{11}{4}$. To find the number of ways to form a korfball team with four men and four women from a group of 7 men and 11 women, the number of ways to choose 4 men out of 7 men and the number of ways to choose 4 women out of 11 women. The total number of ways to form a korfball team can be obtained by multiplying these two quantities.

□

18. **Combinations: Binomial Coefficient 2** (10 points)
Give a combinatorial argument[2] to prove the following Binomial Identity. $k\binom{n}{k} = n\binom{n-1}{k-1}$.

Proof:

■

19. **Combinatorial Proof: Binomial Sum (Vandermonde's Identity)** (10 points)
Let m and w be positive integers. Give a combinatorial argument (see footnote 2) to prove that for integers $k \geq 0$, $\sum_{j=0}^{k} \binom{m}{j}\binom{w}{k-j} = \binom{m+w}{k}$.

Proof:

■

20. **Permutations of Multisets (with repetition): Multinomial Coefficient** (10 points)
Determine the coefficient on $x^{15}y^{120}z^{25}$ in $(2x + 3y^2 + z)^{100}$.

Answer: The coefficient of $x^{15}y^{120}z^{25}$ is $\binom{100}{15,60,25} \cdot 2^{15} \cdot 3^{60}$. To see this, use the Multinomial Theorem to find the multinomial coefficient of $a^{15}b^{60}c^{25}$ in the expansion of $(a + b + c)^{100}$, then substitute $a = 2x$, $b = 3y^2$, and $c = z$ to get the answer.

□

21. **Permutations of Multisets (with repetition): Polynomial Coefficient** (10 points)
Determine the coefficient on $x^{12}y^{24}$ in $(x^3 + 2xy^2 + y + 3)^{18}$. (Be careful, as x and y now appear in multiple terms!)

Answer:

□

[2] A **combinatorial proof** is a method of proving a mathematical identity by showing that both sides of the equation count the same set of objects.

22. **Inclusion-Exclusion Principle 1** (10 points)
Suppose we are making license plates of the form $l_1 l_2 l_3$-$d_1 d_2 d_3$ where l_1, l_2, l_3 are capital letters in the English alphabet and d_1, d_2, d_3 are decimal digits (i.e., elements of the set $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$) subject to the restriction that at least one digit is nonzero and at least one letter is K. How many license plates can we make?

Answer: $(26^3 - 25^3) \cdot (10^3 - 1)$. The total number of choices for $l_1 l_2 l_3$ is $26^3 - 25^3$ because there are 26^3 total strings and 25^3 of them contain no K. The total number of choices for $d_1 d_2 d_3$ is $10^3 - 1$ because there are 10^3 total strings and 1 of them contains only zeroes.

□

23. **Inclusion-Exclusion Principle 2** (10 points)
How many integer solutions are there to the equation $x_1 + x_2 + x_3 + x_4 = 132$ provided that $x_1 > 0$, and $x_2, x_3, x_4 \geq 0$? What if we add the restriction that $x_4 < 17$?

Answer: $(x_1 - 1) + x_2 + x_3 + x_4 = 132 - 1 = 131$ $(134, 3)$. With the restriction that $x_4 < 17$, $(134, 3) - (117, 3)$ because we are subtracting out the cases where $x_4 \geq 17$.

□

24. **Inclusion-Exclusion Principle 3** (10 points)
How many integer-valued solutions are there to each of the following equations and inequalities?

a. $(x_1 - 1) + (x_2 - 1) + (x_3 - 1) + (x_4 - 1) + (x_5 - 1) = 58$, all $(x_i - 1) \geq 0$

Answer: By using combinations with repetition, we can solve each question as follows.
$(58 + 5 - 1, 58) = (62, 4)$

□

b. $x_1 + x_2 + x_3 + x_4 + x_5 = 63$, all $x_i \geq 0$

Answer: $(63 + 5 - 1, 63) = (67, 4)$

□

c. $x_1 + x_2 + x_3 + x_4 + x_5 + x_6 = 63$, all $x_i \geq 0$

Answer: $(63 + 6 - 1, 63) = (68, 5)$

□

d. $x_1 + (x_2 - 10) + x_3 + x_4 + x_5 = 53$, all $x_i \geq 0$, $(x_2 - 10) \geq 0$

Answer: $(53 + 5 - 1, 53) = (57, 4)$

□

e. $x_1 + x_2 + x_3 + x_4 + x_5 = 63$, all $x_i \geq 0$, $x_2 \leq 9$

Answer: Since $x_2 \leq 9$ is a negation of $x_2 \geq 10$ since all x_i's are integers, it is trivial to find the relationship of sub-questions, (b), (d), and (e). Let B be a set of integer pairs satisfying the equation or inequality (b), D be a set of integer pairs satisfying (d), and E be a set of integer pairs satisfying (e). Clearly, $D \cap E = \emptyset$ and $D \cup E = B$, which implies that $|E| = |B| - |D|$. Therefore, $(67, 4) - (57, 4)$

□

25. **Inclusion-Exclusion Principle 4** (10 points)
How many integer-valued solutions are there to each of the following equations and inequalities?

a. $x_1 + x_2 + x_3 + x_4 + x_5 = 63$, all $x_i > 0$

Answer:

☐

b. $x_1 + x_2 + x_3 + x_4 + x_5 = 63$, all $x_i \geq 0$

Answer:

☐

c. $x_1 + x_2 + x_3 + x_4 + x_5 \leq 63$, all $x_i \geq 0$

Answer:

☐

d. $x_1 + x_2 + x_3 + x_4 + x_5 = 63$, all $x_i \geq 0$, $x_2 \geq 10$

Answer:

☐

e. $x_1 + x_2 + x_3 + x_4 + x_5 = 63$, all $x_i \geq 0$, $x_2 \leq 9$

Answer:

☐

26. **Inclusion-Exclusion Principle 5** (10 points)
A teacher has 450 identical pieces of candy. He wants to distribute them to his class of 65 students, although he is willing to take some leftover candy home. (He does not insist on taking any candy home, however.) The student who won a contest in the last class is to receive at least 10 pieces of candy as a reward. Of the remaining students, 34 of them insist on receiving at least one piece of candy, while the remaining 30 students are willing to receive no candy.

a. In how many ways can he distribute the candy?

Answer: $\binom{471}{65}$

☐

b. In how many ways can he distribute the candy if, in addition to the conditions above, one of his students is diabetic and can receive at most 7 pieces of candy? (This student is one of the 34 who insist on receiving at least one piece of candy.)

Answer: $\binom{471}{65} - \binom{464}{65}$

☐

27. **Lattice Paths 1** (10 points)
How many lattice paths are there from (0,0) to (10,12) ? How many lattice paths are there from (3,5) to (10,12) ?

Answer: $(14, 7)$. We must walk 14 steps, 7 of which must go right, and 7 of which must go up. It is the same as the number of ways to flip a coin 14 times and get 7 heads and 7 tails.

□

28. **Lattice Paths 2** (10 points)
How many lattice paths from (0,0) to (10,12) that pass through (3,5) ?

Answer:

□

29. **Lattice Paths 3** (10 points)
How many lattice paths from (0,0) to (17,12) are there that pass through (7,6) and (12,9) ?

Answer: $(13, 6) \cdot (8, 5) \cdot (8, 3)$. There are $(13, 6)$ paths between the first two points, $(8, 5)$ paths between the second point and the third point, and $(8, 3)$ paths between the third point and the last point.

□

30. **Lattice Paths 4** (10 points)
How many lattice paths from (0,0) to (14,73) are there that do not pass through (6,37) ?

Answer: $(87, 14) - (43, 6) \cdot (44, 8)$. Let A be a set of lattice paths from (0,0) to (14,73), B be a set of lattice paths from (0,0) to (14,73) passing through (6,37), and C be a set of lattice paths from (0,0) to (14,73) NOT passing through (6,37). Clearly, $B \cap C = \emptyset$ and $B \cup C = A$, which implies that $|C| = |A| - |B|$. Therefore, $(87, 14) - (43, 6) \cdot (44, 8)$, that is, $(87, 14) - (43, 6) \cdot (44, 8)$

□

31. **Lattice Paths 5** (10 points)
A small-town bank robber is driving his getaway car from the bank he just robbed to his hideout. The bank is at the intersection of 1$^{\text{st}}$ Street and 1$^{\text{st}}$ Avenue. He needs to return to his hideout at the intersection of 7$^{\text{th}}$ Street and 5$^{\text{th}}$ Avenue. However, one of his lookouts has reported that the town's one police officer is parked at the intersection of 4$^{\text{th}}$ Street and 4$^{\text{th}}$ Avenue. Assuming that the bank robber does not want to get arrested and drives only on streets and avenues, in how many ways can he safely return to his hideout? (Streets and avenues are uniformly spaced and numbered consecutively in this small town.)

Answer: $(10, 4) - (6, 3)(4, 3)$. There are $(10, 4)$ paths from the bank to the hideout. Of those paths, $(6, 3)(4, 3)$ go through the police officer. Subtracting these bad paths out gets us the answer.

□